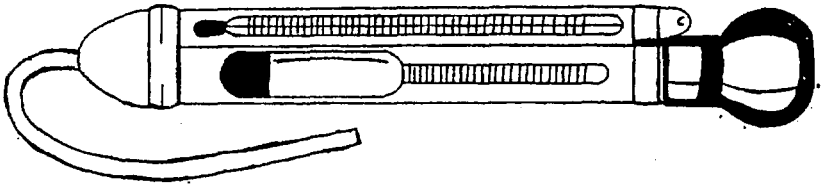
FED. SUP CLASS
6630



PART NUMBER	NUMBER OF HYDROMETERS	THERMOMETER RANGE (°F)	NOMINAL O/A LENGTH OF TESTER (IN.)
MS36545-1	1	-30 to 220	33
MS36545-2	1	-30 to 220	30
MS36545-3	4	-60 to 160	34

MATERIAL: Hydrometers, tubes, and barrels shall be of borosilicate glass in accordance with Ⓐ ASTM E 438, Type I, Class A.

Bulb and tip shall be of elastomer which will not deteriorate under continued contact with antifreeze solutions.

Ⓐ Scales shall be printed on white plastic or No. 1 white sulfite paper. Markings and inscriptions shall be of permanent black or red marking material, such as India ink.

Ballast shall be of lead shot, mercury, or fused metal alloy.

DESIGN: Tester shall consist of a bulb, a barrel, a tip, a glass tube, a thermometer, and a hydrometer.

Bulb shall be capable of drawing solution into the barrel and retaining it when the bulb is released. It shall be removable so the hydrometers can be interchanged.

Barrels and glass tubes shall be cylindrical with straight sides, and shall be fastened together in parallel. The barrel shall house the hydrometer and the tube shall house the thermometer and freezing temperature chart.

Inlet tube shall be fastened to the lower end of the barrel and glass tube for inserting into the solution.

Plug shall be placed in the upper end of the barrel containing the hydrometer to prevent the hydrometer from entering the bulb.

Hydrometers shall be of the constant-mass, variable displacement type and shall have a stem and body.

Ballast shall be securely contained in the lower portion of the hydrometer body.

Hydrometer stem shall have a scale calibrated in specific gravity units to relate directly to the freezing temperature chart.

Freezing temperature chart shall have a vertical column of freezing temperature values. Temperatures above 0°F shall be printed in black and those below 0°F shall be printed in red with a minus sign.

MS36545-3 tester shall be supplied with a chart to determine the number of quarts of antifreeze necessary to protect known quantities of solutions from freezing at a predetermined temperature. It shall cover a solution capacity range from 8 to 60 quarts in 4 quart increments.

APPROVED 7 OCT 1968 REVISED Ⓐ 18 JUN 1986

P.A.	EA	INTERNATIONAL INTEREST	TITLE	MILITARY STANDARD
Other Cust	MS 99		TESTER, ANTIFREEZE SOLUTIONS	MS36545
Procurement Specifications: GG-T-250		SUPERSEDES:		SHEET 1 OF 2

DD FORM 1 MAY 73 672 (COORDINATED) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE PROJECT NUMBER 6630-0438-3 REVISION

USER ACTIVITIES: ARMY - AV, AR, ER, ME
NAVY - CG, MC, OS

REVIEWER ACTIVITIES: NAVY - YD

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PURPOSE: MS36545-1 tester shall be used for determining the freezing temperature of inhibited ethylene glycol solutions and inhibited methanol solutions.

MS36545-2 tester shall be used for determining the freezing temperature of inhibited ethylene glycol solutions.

MS36545-3 tester shall be used for determining the freezing temperature of inhibited ethylene glycol solutions and ethyl alcohol solutions.

NOTES:
1. MS36545-3 shall have two hydrometers for testing ethylene glycol and two hydrometers for testing ethyl alcohol.
2. Items differing from this Military Standard by minor variations in design or dimensions are considered standard under this document, provided such variations do not affect performance and are not in conflict with the material requirements as specified herein.
3. For design feature purposes, this standard takes precedence over procurement document referenced herein.
4. Referenced documents shall be of the issue in effect on date of invitations for bid.

USER ACTIVITIES:

REVIEWER ACTIVITIES:

REVISED

APPROVED

This military standard is approved for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

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DD FORM 1 MAY 73 672 (COORDINATED) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE

PROJECT NUMBER 6630-0438-3
REVISION

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